

# ON TRIPLE QUADRATIC RESIDUE SYMBOLS IN REAL QUADRATIC FIELDS

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**ABSTRACT.** We introduce triple quadratic residue symbols  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  for certain finite primes  $\mathfrak{p}_i$ 's of a real quadratic field  $k$  with trivial narrow class group. For this, we determine a presentation of the Galois group of the maximal pro-2 Galois extension over  $k$  unramified outside  $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3$  and infinite primes, from which we derive mod 2 arithmetic triple Milnor invariants  $\mu_2(123)$  yielding the triple symbol  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] = (-1)^{\mu_2(123)}$ . Our symbols  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  describes the decomposition law of  $\mathfrak{p}_3$  in a certain dihedral extension  $K$  over  $k$  of degree 8, determined by  $\mathfrak{p}_1, \mathfrak{p}_2$ . The field  $K$  and our symbols  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  are generalizations over real quadratic fields of Rédei's dihedral extension of  $\mathbb{Q}$  and Rédei's triple symbol of rational primes. We give examples of Rédei type extensions  $K$  over real quadratic fields. We also give a cohomological interpretation of our symbols in terms of Massey products.

## INTRODUCTION

The purpose of this paper is to introduce the triple quadratic residue symbol  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  for certain primes  $\mathfrak{p}_i$ 's of real quadratic fields and investigate its properties. Our symbols is a triplication of quadratic residue symbol in real quadratic fields, and is also a generalization of Rédei's triple symbol over the rational number field  $\mathbb{Q}$  to real quadratic fields.

The study of triple symbols in arithmetic goes back to the work of Rédei in 1939 ([R]), which intended to generalize the Legendre symbol and Gauss' genus theory for quadratic fields. For certain prime numbers  $p_1, p_2$  and  $p_3$ , Rédei's triple symbol  $[p_1, p_2, p_3] \in \{\pm 1\}$  describes the decomposition law of  $p_3$  in the dihedral extension of degree 8 over  $\mathbb{Q}$ , determined by  $p_1$  and  $p_2$ . Some variants of the Rédei symbol have also been studied in [Fr], [Fu], [Su] among others. In the late 1990s, Morishita interpreted the Rédei symbol as a mod 2 arithmetic analogue of the triple linking number (Milnor invariant) of a link  $L$  in the 3-sphere  $S^3$  ([Mil], [Tu]) from the viewpoint of arithmetic topology ([Mo4]), especially from the analogy between Koch's theorem on a group presentation of the Galois group  $\pi_1^{\text{pro-2}}(\text{Spec}(\mathbb{Z}) \setminus S)$  of the maximal pro-2 extension over  $\mathbb{Q}$  unramified outside a finite set  $S$  of primes and Milnor's theorem on a group presentation of a link group  $\pi_1(S^3 \setminus L)$ , and introduced the multiple quadratic residue symbols  $[p_1, \dots, p_r]$  for certain prime numbers  $p_i$ 's as mod 2 arithmetic analogues of Milnor invariants of a link ([Mo1], [Mo2], [Mo3]). He also gave a description of  $[p_1, \dots, p_r]$  in terms of the Massey product of the étale cohomology of  $\text{Spec}(\mathbb{Z}) \setminus \{(p_1), \dots, (p_r)\}$  ([Mo3]).

Now it remains an important problem to generalize the multiple quadratic residue symbols in  $\mathbb{Q}$  to the multiple power residue symbols for primes in a number field. We may note that Turaev ([Tu]) extended Milnor's theorem on a link group to an integral homology 3-sphere  $M$  and defined Milnor's invariants of a link in  $M$ . Since the ideal class group is regarded as an analogue of the 1st homology group of a 3-manifold in arithmetic topology, it is natural to expect that we may extend Koch's presentation of the pro-2 Galois group over  $\mathbb{Q}$  to the Galois group  $G_{k,S}(l)$  of the maximal pro- $l$  extension over a number field  $k$  unramified outside a finite set  $S$  of primes, when  $k$  has the (narrow) class number one and  $k$  contains the  $l$ -th roots of unity ( $l$  being a prime number), and that we may derive multiple  $l$ -th power residue symbols from

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$G_{k,S}(l)$ . However, there are a cohomological obstruction to determine a group presentation of  $G_{k,S}(l)$  and the difficulty arising from the unit group of  $k$ , in order to define arithmetic Milnor invariants. So far there is the only case  $k = \mathbb{Q}(\sqrt{-3})$  and  $l = 3$ , for which triple cubic residue symbols are defined by using the Galois group  $G_{\mathbb{Q}(\sqrt{-3}),S}(3)$  ([AMM]). For other approaches to triple symbols, we refer to the recent works [E] and [KM].

In this paper, we resolve the difficulty arising from the cohomological obstruction and the unit group for a real quadratic field  $k$  and determine a presentation of the Galois group  $G_S(2)$  of the maximal pro-2 extension of  $k$  unramified outside a finite set of finite primes  $S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_s\}$  and the infinite primes, when  $k$  has the trivial narrow class group. From this we are able to derive the triple quadratic residue symbol  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] = \pm 1$  for certain finite primes of  $k$ . To be precise, fix a prime number  $p \equiv 1 \pmod{4}$  and let  $k$  be a real quadratic field  $\mathbb{Q}(\sqrt{p})$  whose narrow ideal class group is trivial, and let  $\varepsilon$  be the fundamental unit of  $k$ . Let  $S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_s\}$  be a finite set of finite primes of  $k$  and let  $\infty_1, \infty_2$  denotes the infinite primes of  $k$ , where  $N\mathfrak{p}_i \equiv 1 \pmod{4}$  and the corresponding embeddings  $\iota_{\infty_j} : k \hookrightarrow \mathbb{R}$  are defined by  $\iota_{\infty_1}(a + b\sqrt{p}) = a + b\sqrt{p}$ ,  $\iota_{\infty_2}(a + b\sqrt{p}) = a - b\sqrt{p}$  for  $a, b \in \mathbb{Q}$ . In Section 1, we showed as our first result that the pro-2 Galois group  $G_S(2)$  has the following presentation

$$G_S(2) = \langle x_1, \dots, x_s \mid x_1^{N\mathfrak{p}_1-1}[x_1, y_1] = \dots = x_s^{N\mathfrak{p}_s-1}[x_s, y_s] = 1, x_{\infty_2}^2 = 1 \rangle, \quad (0.1)$$

where the word  $x_i$  represents a monodromy  $\tau_i$  over  $\mathfrak{p}_i$  in  $k_S(2)/k$  and  $y_i$  denotes the free pro-2 word of  $x_1, \dots, x_s$  which represents a Frobenius automorphism  $\sigma_i$  over  $\mathfrak{p}_i$  in  $k_S(2)/k$ , and  $x_{\infty_2}$  denotes the word which represents the monodromy over a real prime  $\infty_2$  (see Theorem 1.3). We note that the presentation (0.1) looks similar to that of a link group ([Mil], [Tu]) by the analogy between a monodromy (resp. Frobenius automorphism) over a prime and a meridian (resp. longitude) of a knot ([Mo4, Chapter 8]).

In Section 2, we define the triple quadratic residue symbol  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  using group presentation obtained in Section 1, and we also confirm that this definition is a generalization of the quadratic residue symbol. To be precise, let us consider the case that  $S = \{\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3\}$  with  $\mathfrak{p}_i = (\pi_i)$  such that

$$\left(\frac{\pi_i}{\pi_j}\right) = 1 \text{ for } 1 \leq i \neq j \leq 3 \text{ and } \left(\frac{\varepsilon}{\pi_i}\right) = 1 \text{ for } 1 \leq i \leq 3,$$

where  $\left(\frac{\cdot}{\cdot}\right)$  denotes the quadratic residue symbol in  $k$ . Then, following the method by Morishita ([Mo1], [Mo2], [Mo3]), we obtain the well-defined mod 2 Milnor invariant  $\mu_2(123)$  by using the presentation (0.1) and hence the triple quadratic residue symbol

$$[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] := (-1)^{\mu_2(123)}.$$

By the definition of  $\mu_2(123)$ , the triple symbol  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  describes the decomposition law of  $\mathfrak{p}_3$  in a dihedral extension  $K$  of degree 8 over  $k$ , unramified outside  $S$  and the infinite primes with ramification index of  $\mathfrak{p}_i$  being 2. We call such an extension  $K$  a *Rédei type  $D_8$ -extension* over  $k$ . Actually we show that Rédei type extension  $K$  exists uniquely for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  (see Theorem 2.6).

In Section 3, we construct Rédei type  $D_8$ -extension  $K$  under several conditions, following the method by Rédei ([A], [R]). To be precise, let  $k = \mathbb{Q}(\sqrt{p})$  be a real quadratic field and we assume  $p \equiv 5 \pmod{8}$ . Let  $\mathfrak{p}_1 = (\pi_1), \mathfrak{p}_2 = (\pi_2)$  are prime ideals of  $\mathcal{O}_k$  such that  $\pi_i \equiv 1 \pmod{4\mathcal{O}_k}$  and  $\pi_i > 0$  ( $1 \leq i \leq 2$ ). Then, there is a non-zero  $\mathcal{O}_k$  solution  $(x, y, z)$  of  $x^2 - \pi_1 y^2 - \pi_2 z^2 = 0$  such that  $\gcd(x, y, z) = 1$ ,  $y \in 2\mathcal{O}_k$ . Furthermore, if  $x - y \equiv 1 \pmod{4\mathcal{O}_k}$ , then the Rédei type  $D_8$ -extension  $K$  is given by

$$K = k(\sqrt{\pi_1}, \sqrt{\pi_2}, \sqrt{x + y\sqrt{\pi_1}}).$$

We also give some numerical examples of Rédei type  $D_8$ -extensions over real quadratic fields. Moreover, we give the first example of a *Borromean primes*  $\mathfrak{p}_1 = (\pi_1), \mathfrak{p}_2 = (\pi_2), \mathfrak{p}_3 = (\pi_3)$  in  $\mathbb{Q}(\sqrt{5})$ , which satisfies  $\left(\frac{\pi_i}{\pi_j}\right) = 1$  ( $i \neq j$ ) and  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] = -1$ .

Finally, in Section 4, following [Mo3], we give a cohomological interpretation of mod 2 Milnor invariant and the triple quadratic residue symbol from the perspective of Massey products.

**Notations.** For a set  $A$ ,  $\#A$  denotes the cardinality of  $A$ .

For a finite algebraic number field  $F$ ,  $\mathcal{O}_F$  denotes the ring of integers of  $F$ ,  $H_F$  denotes the ideal class group of  $F$ , and  $H_F^+$  denotes the narrow ideal class group of  $F$ .

For an ideal  $\mathfrak{a}$  of  $\mathcal{O}_F$ ,  $N\mathfrak{a} := \#(\mathcal{O}_k/\mathfrak{a})$ .

For a group  $G$  and  $x, y \in G$ ,  $[x, y] := xyx^{-1}y^{-1}$ .

For a topological group  $G$  and subsets  $X, Y$  of  $G$ ,  $[X, Y]$  denotes the closed subgroup of  $G$  generated by  $[x, y]$  for  $x \in X, y \in Y$ .

## 1. MAXIMAL PRO-2 GALOIS GROUPS OF REAL QUADRATIC FIELDS WITH RESTRICTED RAMIFICATION

In this section, we study a presentation of the Galois group  $G_S(2)$  of the maximal pro-2 extension over a real quadratic field  $k$ , which is unramified outside a finite set  $S$  of primes of  $k$ . For this, we apply theorems due to Höchsmann and Koch ([Ko]) and we assume that the narrow class group of  $k$  is trivial and a certain cohomological obstruction  $B_S$  vanishes, in order to give a precise minimal generators and their relations. We also give a condition for  $B_S = \{1\}$  in terms of the fundamental unit of  $k$ .

Let  $k$  be a real quadratic field and  $\mathcal{O}_k$  be the ring of integers of  $k$ . We recall that the unit group  $\mathcal{O}_k^\times$  is given by

$$\mathcal{O}_k^\times = \{\pm \varepsilon^n \mid n \in \mathbb{Z}\},$$

where  $\varepsilon$  is the fundamental unit of  $k$ . Let  $S$  be a finite set of finite primes of  $k$ ,

$$S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_s\}, \quad \#S = s.$$

We assume that  $\mathfrak{p}_i$ 's are not lying over 2, that is,  $N\mathfrak{p}_i \equiv 1 \pmod{2}$  for each  $i$ . Let  $\bar{S}$  be the union

$$\bar{S} = S \cup S_{\mathbb{R}},$$

where  $S_{\mathbb{R}} = \{\infty_1, \infty_2\}$  is the set of real primes of  $k$ . For  $\mathfrak{p}_i \in S$ , let  $k_{\mathfrak{p}_i}$  be the completion of  $k$  at  $\mathfrak{p}_i$  and let  $\pi_i$  be a prime element of  $k_{\mathfrak{p}_i}$ . Each primes  $\mathfrak{p} \in \bar{S}$  defines an embedding  $k$  to  $k_{\mathfrak{p}}$ , which we denote by  $\iota_{\mathfrak{p}}$ . In particular, for real primes, we denote by  $\iota_{\infty_1}$  the embedding into  $\mathbb{R}$  determined from the identity map and the conjugate map by  $\iota_{\infty_2}$ . Let  $k_S(2)$  denote the maximal pro-2 extension of  $k$ , unramified outside  $\bar{S}$  and we set  $G_S(2) := \text{Gal}(k_S(2)/k)$ .

Now we describe the structure of the pro-2 group  $G_S(2)$  in a certain unobstructed case. For this, we first recall a result due to Iwasawa on the local Galois group ([Ko, Satz 10.2]). We fix an algebraic closure  $\bar{k}_{\mathfrak{p}_i}$  of  $k_{\mathfrak{p}_i}$  and an embedding  $\bar{k} \hookrightarrow \bar{k}_{\mathfrak{p}_i}$ . Let  $k_{\mathfrak{p}_i}(2)$  denote the maximal pro-2 extension of  $k_{\mathfrak{p}_i}$  in  $\bar{k}_{\mathfrak{p}_i}$  and  $G_{k_{\mathfrak{p}_i}}(2)$  denote the Galois group of  $k_{\mathfrak{p}_i}(2)$  over  $k_{\mathfrak{p}_i}$ . In this case, the following identity is known:

$$k_{\mathfrak{p}_i}(2) = k_{\mathfrak{p}_i}(\zeta_{2^n}, \sqrt[n]{\pi_i} \mid n \geq 1),$$

where  $\zeta_{2^n}$  denotes a primitive  $2^n$ -th root of unity in  $\bar{k}$  such that  $(\zeta_{2^a})^{2^b} = \zeta_{2^{a-b}}$  for all  $a \geq b$ . The local Galois group  $G_{k_{\mathfrak{p}_i}}(2)$  is topologically generated by the monodromy  $\tau_i$  and (a lift of) the Frobenius automorphism  $\sigma_i$  which are defined by

$$\begin{aligned} \tau_i(\zeta_{2^n}) &:= \zeta_{2^n}, & \tau_i(\sqrt[n]{\pi_i}) &:= \zeta_{2^n} \sqrt[n]{\pi_i}, \\ \sigma_i(\zeta_{2^n}) &:= \zeta_{2^n}^{N\mathfrak{p}_i}, & \sigma_i(\sqrt[n]{\pi_i}) &:= \sqrt[n]{\pi_i}, \end{aligned}$$

which are subject to the relation

$$\tau_i^{N\mathfrak{p}_i-1}[\tau_i, \sigma_i] = 1. \tag{1.1}$$

For  $\mathfrak{p} \in S_{\mathbb{R}}$ , we have  $k_{\mathfrak{p}} = \mathbb{R}$ ,  $k_{\mathfrak{p}}(2) = \mathbb{C}$ , and

$$G_{k_{\mathfrak{p}}}(2) = \langle \tau \mid \tau^2 = 1 \rangle. \tag{1.2}$$

For each  $\mathfrak{p} \in \overline{S}$ , the fixed embedding  $\bar{k} \hookrightarrow \bar{k}_{\mathfrak{p}}$  induces an embedding  $k_S(2) \hookrightarrow k_{\mathfrak{p}}(2)$ , so a prime  $\mathfrak{P}_i$  of  $k_S(2)$  lying over  $\mathfrak{p}_i$  ( $1 \leq i \leq s$ ). We denote by the same letters  $\tau_i$  and  $\sigma_i$  the images of  $\tau_i$  and  $\sigma_i$  under  $\varphi_{\mathfrak{p}}$ , respectively, under the homomorphism

$$\varphi_{\mathfrak{p}} : G_{k_{\mathfrak{p}}}(2) \longrightarrow G_S(2)$$

induced by the embedding  $k_S(2) \hookrightarrow k_{\mathfrak{p}}(2)$ . Then  $\tau_i$  is a topological generator of the inertia group of  $\mathfrak{P}_i$  and  $\sigma_i$  is a lift of the Frobenius automorphism of the maximal subextension of  $k_S(2)/k$  in which  $\mathfrak{P}_i$  is unramified. We call simply  $\tau_i$  a *monodromy over  $\mathfrak{p}_i$*  in  $k_S(2)/k$  and  $\sigma_i$  a *Frobenius automorphism over  $\mathfrak{p}_i$*  in  $k_S(2)/k$ . We note that the restriction of  $\tau_i$  to the maximal Abelian subextension  $k_S(2)^{\text{ab}}$  in  $k_S(2)/k$ , denoted by  $\bar{\tau}_i := \tau_i|_{k_S(2)^{\text{ab}}}$ , is given by the Artin symbol

$$\bar{\tau}_i = (\eta_i, k_S(2)^{\text{ab}}/k),$$

for an idèle  $\eta_i$  of  $k$  whose  $\mathfrak{p}_i$ -component is a primitive  $(N\mathfrak{p}_i - 1)$ -th root of unity in  $k_{\mathfrak{p}_i}^{\times}$  and all of the other components are 1. Now we assume that the 2-class group  $H_k(2)$  of  $k$  is trivial. Then, by class field theory and Burnside's basis theorem ([Ko, Satz 4.10]), the monodromies  $\tau_1, \dots, \tau_s, \tau_{\infty_1}, \tau_{\infty_2}$  generate topologically the global Galois group  $G_S(2)$ . However, they may not be a minimal set of generators in general. In fact, Shafarevich's theorem ([Ko, Satz 11.8]) tells us that the minimal number  $d(G_S(2))$  of generators of  $G_S(2)$  is given by

$$d(G_S(2)) = s + \dim_{\mathbb{F}_2} B_S, \quad (1.3)$$

where the obstruction  $B_S$  is defined as follows: We define the subgroup  $V_S$  of  $k^{\times}$  by

$$V_S := \{a \in k^{\times} \mid a\mathcal{O}_k = \mathfrak{a}^2, a \in (k_{\mathfrak{p}}^{\times})^2 (\forall \mathfrak{p} \in \overline{S})\},$$

where  $\mathfrak{a}$  is a fractional ideal of  $\mathcal{O}_k$ , and then  $B_S$  is defined by

$$B_S := V_S / (k^{\times})^2,$$

which is an elementary Abelian 2-group. By Shafarevich's formula (1.3) on  $d(G_S(2))$ ,  $2 - \dim_{\mathbb{F}_2} B_S$  generators among  $\tau_i$ 's have to be removed in order to get a minimal generators of  $G_S(2)$ .

Now we show a necessary and sufficient condition for  $B_S$  to vanish by using the fundamental unit  $\varepsilon$  of  $k$ . For  $a \in k^{\times}$  and a finite prime  $\mathfrak{p}$  of  $k$  such that  $2, a$  and  $\mathfrak{p}$  are relatively prime,  $\left(\frac{a}{\mathfrak{p}}\right)$  denotes the quadratic residue symbol in  $k$ . For a real prime  $\infty$  of  $k$ ,  $\left(\frac{a}{\infty}\right)$  is defined by 1 (resp.  $-1$ ) if  $\iota_{\infty}(a) > 0$  (resp. if  $\iota_{\infty}(a) < 0$ ).

**Proposition 1.1.** (1) *We assume that the ideal class group  $H_k$  of  $k$  is trivial. Then, the following conditions are equivalent.*

(i)  $B_S = \{1\}$ .

(ii) *There is a prime  $\mathfrak{p} \in \overline{S}$  such that  $\left(\frac{\varepsilon}{\mathfrak{p}}\right) = -1$ .*

(2) *We assume that the narrow ideal class group  $H_k^+$  of  $k$  is trivial. Then we have  $\left(\frac{\varepsilon}{\infty_2}\right) = -1$  and  $B_S = \{1\}$ .*

**Proof.** (1) Suppose that  $V_S = (k^{\times})^2$ . Since  $\varepsilon$  is not square in  $k^{\times}$ ,  $\varepsilon$  is not included in  $V_S$ . Since  $\varepsilon\mathcal{O}_k = (1)^2$ , there exists  $\mathfrak{p} \in \overline{S}$  such that  $\varepsilon \notin (k_{\mathfrak{p}}^{\times})^2$ . Consequently, we have  $\left(\frac{\varepsilon}{\mathfrak{p}}\right) = -1$  for some prime  $\mathfrak{p} \in \overline{S}$ .

Conversely, we assume that there is a prime  $\mathfrak{p} \in \overline{S}$  such that  $\left(\frac{\varepsilon}{\mathfrak{p}}\right) = -1$ . Choose any  $\alpha \in V_S$ . By the definition of  $V_S$ , there is an ideal  $\mathfrak{a}$  satisfying  $\alpha\mathcal{O}_k = \mathfrak{a}^2$ . Since the class number of  $k$  is one,  $\mathfrak{a}$  is generated by one element  $a \in k^{\times}$ . So,  $\alpha$  has the form  $\alpha = u \cdot a^2$  ( $u \in \mathcal{O}_k^{\times}$ ). Because of the structure of  $\mathcal{O}_k^{\times}/(\mathcal{O}_k^{\times})^2$ , we have  $u = u' \cdot v^2$  ( $u' \in \{1, -1, \varepsilon, -\varepsilon\}$ ,  $v \in \mathcal{O}_k^{\times}$ ). Then replacing  $v \cdot a$  by  $a$ ,  $\alpha$  has the form  $u' \cdot a^2$  ( $u' \in \{1, -1, \varepsilon, -\varepsilon\}$ ). So we obtain  $u' = \alpha \cdot (a^{-1})^2 \in V_S$ . On the other hand,  $-1, -\varepsilon \notin V_S$  because  $\overline{S}$  has a real prime  $\infty_1$ . In addition,  $\varepsilon$  is not included in  $V_S$  by the assumption. Consequently, we have  $u' = 1$  and  $\alpha = a^2 \in (k^{\times})^2$ .

(2) Since the narrow ideal class group  $H_k^+$  of  $k$  is trivial,  $N_{k/\mathbb{Q}}(\varepsilon) = \varepsilon \cdot \iota_{\infty_2}(\varepsilon) = -1$ . Therefore we have  $\iota_{\infty_2}(\varepsilon) = -1$  and  $\left(\frac{\varepsilon}{\infty_2}\right) = -1$ . By (1), we have  $B_S = \{1\}$ .  $\square$

By using Proposition 1.1, the minimal set of generators of  $G_S(2)$  can be determined as follows. Let  $\Phi(G_S(2))$  be the Frattini subgroup of  $G_S(2)$ , namely,  $\Phi(G_S(2)) = G_S(2)^2[G_S(2), G_S(2)]$ .

**Proposition 1.2.** *Notations being as above, we assume that the ideal class group  $H_k$  of  $k$  is trivial, and there is a prime  $\mathfrak{p} \in \overline{S} \setminus \{\infty_1\}$  such that  $\left(\frac{\varepsilon}{\mathfrak{p}}\right) = -1$ . Then  $G_S(2)$  is topologically generated by the set of minimal generators  $\{\tau_1, \dots, \tau_s, \tau_{\infty_2}\} \setminus \{\tau\}$ , where  $\tau$  is a monodromy corresponding to  $\mathfrak{p}$ . If the narrow ideal class group  $H_k^+$  of  $k$  is trivial, then we can take  $\tau = \tau_{\infty_2}$ .*

**Proof.** By Burnside's basis theorem, it is sufficient to show that  $\{\tau_1, \dots, \tau_s, \tau_{\infty_2}\} \setminus \{\tau\}$  generates  $G_S(2)/\Phi(G_S(2))$ . For a prime  $\mathfrak{p}$  of  $k$ , let  $\mathcal{O}_{\mathfrak{p}}$  be the ring of  $\mathfrak{p}$ -adic integers of  $k_{\mathfrak{p}}$ . (For the infinite prime  $\mathfrak{p}$ , we set  $\mathcal{O}_{\mathfrak{p}} := k_{\mathfrak{p}}$ .) Let  $J_k$  be the idèle group of  $k$ . Let  $U_k$  be the subgroup of  $J_k$  consisting of unit idèles of  $k$ ,  $U_k := \prod_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^{\times}$ , and let  $U_S$  denote the subgroup of  $U_k$  whose  $\mathfrak{p}$ -component is 1 for any  $\mathfrak{p} \in \overline{S}$ . Employing class field theory, we have the canonical isomorphism of  $\mathbb{F}_2$ -vector spaces

$$J_k/U_S J_k^2 k^{\times} \simeq G_S(2)/\Phi(G_S(2))$$

by the Artin symbol.

Since the ideal class group  $H_k = J_k/U_k k^{\times}$  is trivial and  $B_S = \{1\}$  by Proposition 1.1, we have the following exact sequence of  $\mathbb{F}_2$ -vector spaces (cf. [Ko, (11.11)]).

$$1 \longrightarrow \mathcal{O}_k^{\times}/(\mathcal{O}_k^{\times})^2 \xrightarrow{\delta} \prod_{\mathfrak{p} \in \overline{S}} \mathcal{O}_{\mathfrak{p}_i}^{\times}/(\mathcal{O}_{\mathfrak{p}_i}^{\times})^2 \xrightarrow{\iota} J_k/U_S J_k^2 k^{\times} \longrightarrow 1,$$

where  $\delta$  is the diagonal map and  $\iota$  is induced by the natural inclusion  $\prod_{\mathfrak{p} \in \overline{S}} \mathcal{O}_{\mathfrak{p}_i}^{\times} \hookrightarrow J_k$ . By local and global class field theory,  $\mathcal{O}_{\mathfrak{p}_i}^{\times}/(\mathcal{O}_{\mathfrak{p}_i}^{\times})^2 \simeq \langle \overline{\tau_i} \rangle$  and  $J_k/U_S J_k^2 k^{\times} \simeq G_S(2)/\Phi(G_S(2))$ . So we obtain the following exact sequence.

$$1 \rightarrow \mathcal{O}_k^{\times}/(\mathcal{O}_k^{\times})^2 \xrightarrow{\delta} \langle \overline{\tau_1} \rangle \times \cdots \times \langle \overline{\tau_s} \rangle \times \langle \overline{\tau_{\infty_1}} \rangle \times \langle \overline{\tau_{\infty_2}} \rangle \xrightarrow{\iota} G_S(2)/\Phi(G_S(2)) \rightarrow 1.$$

Therefore,  $\{\iota(\overline{\tau_1}), \dots, \iota(\overline{\tau_s}), \iota(\overline{\tau_{\infty_1}}), \iota(\overline{\tau_{\infty_2}})\}$  forms a system of generators of  $G_S(2)/\Phi(G_S(2))$ . On the other hand, Since  $\mathcal{O}_k^{\times}/(\mathcal{O}_k^{\times})^2 = \{1, -1, \varepsilon, -\varepsilon\} = \langle -1, \varepsilon \rangle$ ,  $\{\delta(-1), \delta(\varepsilon)\}$  is a basis of  $\text{Im}(\delta)$ . Since  $\text{Ker}(\iota) = \text{Im}(\delta)$ ,  $\{\delta(-1), \delta(\varepsilon)\}$  is a basis of  $\text{Ker}(\iota)$ . By expressing  $\delta(-1)$  and  $\delta(\varepsilon)$  as  $\delta(-1) = \overline{\tau_1}^{a_1} \cdots \overline{\tau_{\infty_2}}^{a_{\infty_2}}$ ,  $\delta(\varepsilon) = \overline{\tau_1}^{b_1} \cdots \overline{\tau_{\infty_2}}^{b_{\infty_2}}$  using the basis of  $\langle \overline{\tau_1} \rangle \times \cdots \times \langle \overline{\tau_{\infty_2}} \rangle$ , we obtain the relations

$$\begin{cases} \overline{\tau_1}^{a_1} \cdots \overline{\tau_{\infty_2}}^{a_{\infty_2}} = 1 & (1.2.1) \\ \overline{\tau_1}^{b_1} \cdots \overline{\tau_{\infty_2}}^{b_{\infty_2}} = 1 & (1.2.2) \end{cases}$$

in  $G_S(2)/\Phi(G_S(2))$ . Here, for  $i, j = 1, \dots, s, \infty_1, \infty_2$ , we define

$$a_i = \begin{cases} 1 & \delta(-1) \notin (\mathcal{O}_{\mathfrak{p}_i}^{\times})^2 \\ 0 & \text{otherwise,} \end{cases} \quad b_j = \begin{cases} 1 & \delta(\varepsilon) \notin (\mathcal{O}_{\mathfrak{p}_j}^{\times})^2 \\ 0 & \text{otherwise.} \end{cases}$$

From relation (1.2.1), since  $\delta(-1) = -1 \notin (\mathcal{O}_{\infty_1}^{\times})^2 = (\mathbb{R}^{\times})^2$  we see that it is possible to omit  $\overline{\tau_{\infty_1}}$  from the system of generators of  $G_S(2)/\Phi(G_S(2))$ . Moreover, by definition of  $b_j$  and relation (1.2.2) another omissible element is  $\overline{\tau}$  corresponding to  $\mathfrak{p} \in \overline{S}$  such that  $\left(\frac{\varepsilon}{\mathfrak{p}}\right) = -1$ .

In addition, if the narrow ideal class group of  $k$  is trivial, the same argument as above shows that we can use the relation (1.2.1) (resp. (1.2.2)) to omit  $\tau_{\infty_1}$  (resp.  $\tau_{\infty_2}$ ) from the system of generators. Note that the image of  $\varepsilon$  into  $\mathcal{O}_{\infty_2}^{\times}$  by  $\delta$  is the conjugate of  $\varepsilon$ , and the conjugate of  $\varepsilon$  is a negative number when the narrow ideal class group is trivial.  $\square$

By Proposition 1.1 and Proposition 1.2, the following holds.

**Theorem 1.3.** *Notations being as above, we assume that the narrow ideal class group  $H_k^+$  of  $k$  is trivial. Then the pro-2 group  $G_S(2)$  has the following presentation:*

$$G_S(2) = \langle x_1, \dots, x_s \mid x_1^{\mathbf{Np}_1-1}[x_1, y_1] = \dots = x_s^{\mathbf{Np}_s-1}[x_s, y_s] = 1, x_{\infty_2}^2 = 1 \rangle,$$

where  $x_i$  denotes the word which represents a monodromy  $\tau_i$  over  $\mathfrak{p}_i$  in  $k_S(2)/k$  and  $y_i$  denotes the free pro-2 word of  $x_1, \dots, x_s$  which represents a Frobenius automorphism  $\sigma_i$  over  $\mathfrak{p}_i$  in  $k_S(2)/k$ , and  $x_{\infty_2}$  denotes the word which represents the monodromy over a real prime  $\infty_2$ .

**Proof.** Since the narrow ideal class group  $H_k^+$  of  $k$  is trivial,  $B_S = \{1\}$  by Proposition 1.1 and  $G_S(2)$  is topologically generated by the set of minimal generators  $\{\tau_1, \dots, \tau_s\}$  by Proposition 1.2. Since  $B_S = \{1\}$ , according to [Ko, Satz 11.3], the localization map

$$H^2(G_S(2), \mathbb{F}_2) \longrightarrow \prod_{\mathfrak{p} \in \overline{S}} H^2(G_{k_{\mathfrak{p}}}(2), \mathbb{F}_2)$$

is injective. Moreover, [Ko, Satz 11.4] tells us that the map

$$H^2(G_S(2), \mathbb{F}_2) \longrightarrow \prod_{\mathfrak{p} \in \overline{S} \setminus \{\infty_1\}} H^2(G_{k_{\mathfrak{p}}}(2), \mathbb{F}_2)$$

is also injective. Therefore, by [Ko, Satz 6.14], the relation system of  $G_S(2)$  is generated by the local relation system of  $G_{k_{\mathfrak{p}}}(2)$  ( $\mathfrak{p} \in \overline{S} \setminus \{\infty_1\}$ ). By (1.1) and (1.2), we have presentation of  $G_S(2)$ .  $\square$

The following Proposition 1.4 will be used in Section 2.

**Proposition 1.4.** *Notations being as above, we assume that the narrow ideal class group  $H_k^+$  of  $k$  is trivial. Then, the following conditions are equivalent.*

- (i)  $x_{\infty_2} \in \Phi(G_S(2))$ .
- (ii)  $\left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1$  for all finite primes  $\mathfrak{p}_i \in S$ .

**Proof.** By the proof of Proposition 1.2, we have relations

$$\begin{cases} \overline{\tau_1}^{a_1} \dots \overline{\tau_{\infty_2}}^{a_{\infty_2}} = 1, \\ \overline{\tau_1}^{b_1} \dots \overline{\tau_{\infty_2}}^{b_{\infty_2}} = 1 \end{cases}$$

in  $G_S(2)/\Phi(G_S(2))$ . Here, for  $i, j = 1, \dots, s, \infty_1, \infty_2$ , we define

$$a_i = \begin{cases} 1 & \delta(-1) \notin (\mathcal{O}_{\mathfrak{p}_i}^\times)^2 \\ 0 & \text{otherwise,} \end{cases} \quad b_j = \begin{cases} 1 & \delta(\varepsilon) \notin (\mathcal{O}_{\mathfrak{p}_j}^\times)^2 \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,  $x_{\infty_2} \in \Phi(G_S(2))$  is equivalent to that one of the following conditions hold:

- (iii) for  $i = 1, \dots, s, \infty_1$ ,

$$-1 \in (\mathcal{O}_{\mathfrak{p}_i}^\times)^2 \text{ and } -1 \notin (\mathcal{O}_{\mathfrak{p}_{\infty_2}}^\times)^2,$$

- (iv) for  $j = 1, \dots, s, \infty_1$ ,

$$\varepsilon \in (\mathcal{O}_{\mathfrak{p}_j}^\times)^2 \text{ and } \varepsilon \notin (\mathcal{O}_{\mathfrak{p}_{\infty_2}}^\times)^2.$$

Since  $-1 \notin (\mathcal{O}_{\mathfrak{p}_{\infty_1}}^\times)^2$ , (iii) cannot hold. An equivalent condition for (iv) is that

$$\left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1$$

for all finite primes  $\mathfrak{p}_i \in S$ .  $\square$

## 2. MOD 2 ARITHMETIC MILNOR INVARIANTS AND TRIPLE QUADRATIC RESIDUE SYMBOLS IN REAL QUADRATIC FIELDS

In this section, following Morishita's work ([Mo1], [Mo2], [Mo3]), we introduce mod 2 arithmetic Milnor invariants and triple quadratic residue symbols for certain primes of a real quadratic field, by using the presentation in Theorem 1.3 in Section 1.

Now, let us be back to the arithmetic situation in Section 1 and keep the same notations and assumptions there. Let  $k$  be a real quadratic field. In the following, we assume that the narrow ideal class group  $H_k^+$  of  $k$  is trivial. Let  $S = \{\mathfrak{p}_1, \dots, \mathfrak{p}_s\}$  be a set of  $s$  distinct finite primes of  $k$ , where  $\mathfrak{p}_i$ 's are not lying over 2, and let  $\overline{S} := S \cup S_{\mathbb{R}}$ , when  $S_{\mathbb{R}}$  denotes the set of real primes of  $k$ . We fix a prime element  $\pi_i$  of  $\mathfrak{p}_i$ , namely,  $\mathfrak{p}_i = (\pi_i)$ . By Theorem 1.3, the pro-2 Galois group  $G_S(2)$  has the following presentation:

$$G_S(2) = \langle x_1, \dots, x_s \mid x_1^{\mathbb{N}\mathfrak{p}_1-1}[x_1, y_1] = \dots = x_s^{\mathbb{N}\mathfrak{p}_s-1}[x_s, y_s] = 1, x_{\infty_2}^2 = 1 \rangle,$$

where  $x_i$  denotes the word which represents a monodromy over  $\mathfrak{p}_i$  in  $k_S(2)/k$  and  $y_i$  denotes the free pro-2 word of  $x_1, \dots, x_s$  which represents a Frobenius automorphism over  $\mathfrak{p}_i$  in  $k_S(2)/k$ , and  $x_{\infty_2}$  denotes the word which represents a monodromy over a real prime  $\infty_2$ .

Let  $F_S$  denote the free pro-2 group on the words  $x_1, \dots, x_s$ . Let  $\mathbb{F}_2[[F_S]]$  be the complete group algebra of  $F_S$  over  $\mathbb{F}_2$  and let  $\varepsilon_{\mathbb{F}_2[[F_S]]} : \mathbb{F}_2[[F_S]] \rightarrow \mathbb{F}_2$  be the augmentation homomorphism with the augmentation ideal  $I_{\mathbb{F}_2[[F_S]]} := \text{Ker}(\varepsilon_{\mathbb{F}_2[[F_S]]})$ . Let  $\mathbb{F}_2\langle\langle X_1, \dots, X_s \rangle\rangle$  denote the formal power series algebra over  $\mathbb{F}_2$  in non-commutative variables  $X_1, \dots, X_s$ . For a multiple index  $I = (i_1 \dots i_n)$ ,  $1 \leq i_1, \dots, i_n \leq s$ , we set  $X_I = X_{i_1} \dots X_{i_n}$ . Let

$$M : \mathbb{F}_2[[F_S]] \xrightarrow{\sim} \mathbb{F}_2\langle\langle X_1, \dots, X_s \rangle\rangle,$$

be the mod 2 Magnus isomorphism defined by

$$M(x_i) = 1 + X_i, \quad M(x_i^{-1}) = 1 - X_i + X_i^2 - \dots.$$

Then, for  $f \in F_S$ ,  $M(f)$  has the form (called Mod 2 Magnus expansion)

$$M(f) = 1 + \sum_{|I| \geq 1} \mu_2(I; f) X_I,$$

where  $\mu_2(I; f) \in \mathbb{F}_2$  are the mod 2 Magnus coefficients. In terms of the pro-2 Fox free derivatives  $\frac{\partial}{\partial x_i} : \mathbb{Z}_2[[F_S]] \rightarrow \mathbb{Z}_2[[F_S]]$  over  $\mathbb{Z}_2$ , we can obtain for  $I = (i_1 \dots i_n)$

$$\mu_2(I; f) = \varepsilon_{\mathbb{F}_2[[F_S]]} \left( \frac{\partial^n f}{\partial x_{i_1} \dots \partial x_{i_n}} \bmod 2 \right),$$

where  $\text{mod } 2 : \mathbb{Z}_2[[F_S]] \rightarrow \mathbb{F}_2[[F_S]]$  is the reduction mod 2.

Here, we recall basic properties of mod 2 Magnus coefficients.

**Lemma 2.1.** (i) For  $\alpha, \beta \in \mathbb{F}_2[[F_S]]$  and a multi-index  $I$ ,

$$\mu_2(I; \alpha\beta) = \sum_{I=JK} \mu_2(J; \alpha) \mu_2(K; \beta),$$

where the sum ranges over all pairs  $(J, K)$  of multi-indices such that  $JK = I$ .

(ii) For  $f \in F_S$  and  $q \geq 2$ , we have

$$\mu_2(I; f) = 0 \text{ for } |I| < q \iff f \in F_S^{(q)},$$

where  $F_S^{(q)} := \{x \in F_S \mid g - 1 \in I_{\mathbb{F}_2[[F_S]]}^q\}$ , the  $q$ -th term of the mod 2 Zassenhaus filtration of  $F_S$ .

We note that the set  $\{F_S^{(q)}\}$  forms a fundamental system of neighbourhoods of 1 in  $F_S$  and that  $F_S^{(q)}/F_S^{(q+1)}$  is central in  $F_S/F_S^{(q+1)}$  and  $(F_S^{(q)})^2 \subset F_S^{(2q)}$ .

Now let  $I = (i_1 \cdots i_n)$  be a multi-index with  $1 \leq i_1, \dots, i_n \leq s$ . Let  $I' := (i_1 \cdots i_{n-1})$ . Then the mod 2 arithmetic Milnor number  $\mu_2(I)$  for  $I$  is defined by

$$\mu_2(I) := \mu_2(I'; y_{i_n}).$$

Here we set  $\mu_2(I) := 0$  if  $|I| = 1$ . The mod 2 arithmetic Milnor numbers  $\mu_2(I)$  are not invariants determined by the Galois group  $G_S(2)$ . However, we can show in the following that  $\mu_2(I)$  are invariant of  $G_S(2)$  if  $|I| = 2$  or 3.

**Proposition 2.2.** *Notations being as above, let  $i, j$  be indices between 1 and  $s$ . When  $i = j$ , we have  $\mu_2(ii) = 0$ . When  $i \neq j$ , we have*

$$(-1)^{\mu_2(ij)} = \left( \frac{\pi_i}{\pi_j} \right).$$

**Proof.** By definition of the quadratic residue symbol,

$$\begin{aligned} \left( \frac{\pi_i}{\pi_j} \right) &= \frac{(\pi_j, k_{\mathfrak{p}_j}(\sqrt{\pi_i})/k_{\mathfrak{p}_j})(\sqrt{\pi_i})}{\sqrt{\pi_i}} \\ &= \frac{y_j(\sqrt{\pi_i})}{\sqrt{\pi_i}}. \end{aligned}$$

where  $(\cdot, k_{\mathfrak{p}_j}(\sqrt{\pi_i})/k_{\mathfrak{p}_j}) : k_{\mathfrak{p}_j}^\times \rightarrow \text{Gal}(k_{\mathfrak{p}_j}(\sqrt{\pi_i})/k_{\mathfrak{p}_j})$  is the norm residue symbol. On the other hand, we consider the mod 2 Magnus isomorphism of  $y_j$ ,

$$M(y_j) = 1 + \sum_i \mu_2(ij) X_i + (\text{terms of higher degree}).$$

Therefore  $y_j \equiv \prod_i x_i^{\mu_2(ij)} \pmod{[G_S(2), G_S(2)]}$ . Hence

$$\begin{aligned} \frac{y_j(\sqrt{\pi_i})}{\sqrt{\pi_i}} &= \frac{x_i^{\mu_2(ij)}(\sqrt{\pi_i})}{\sqrt{\pi_i}} \\ &= (-1)^{\mu_2(ij)}. \end{aligned}$$

□

Next, we show that mod 2 arithmetic triple Milnor numbers are invariants determined by  $G_S(2)$ .

**Theorem 2.3.** *Notations being as above, we assume that  $\left( \frac{\varepsilon}{\mathfrak{p}_i} \right) = 1$  for all finite primes  $\mathfrak{p}_i \in S$ ,  $\mu_2(J) = 0$  for any multi-index  $J$  of length  $\leq 2$ , and  $N\mathfrak{p}_i \equiv 1 \pmod{4}$  for all  $\mathfrak{p}_i \in S$ . Then, for a multi-index  $I$  satisfying  $|I| = 3$ ,  $\mu_2(I)$  is an invariant depending on  $G_S(2)$ .*

**Proof.** Let  $I := (i_1, i_2, i_3)$ . We must show that  $\mu_2(I)$  is independent of the choices of a monodromy over  $\mathfrak{p}_i$  and an extension of the Frobenius automorphism over  $\mathfrak{p}_i$ . It suffices to show the following:

- (i)  $\mu_2(I)$  is not changed if  $y_i$  is replaced by a conjugate.
  - (ii)  $\mu_2(I)$  is not changed if  $x_i$  is replaced by a conjugate.
  - (iii)  $\mu_2(I)$  is not changed if  $y_{i_3}$  is multiplied by a conjugate of  $x_i^{N\mathfrak{p}_i-1}[x_i, y_i]$  and  $x_{\infty_2}$ .
- Let  $I' := (i_1, i_2)$ .

**Proof of (i):** Write  $M(y_i) = 1 + \omega_i$  for the Mod 2 Magnus expansion of  $y_i$ ,  $i = 1, 2, 3$ . By the assumption, Lemma 2.1 (ii) and Proposition 2.2, the degree of  $\omega_i \geq 2$ . Then, we have

$$\begin{aligned} M(x_j y_i x_j^{-1}) &= (1 + X_j)(1 + \omega_i)(1 - X_j + X_j^2 - \cdots) \\ &= 1 + \omega_i + (\text{terms involving } X_j \omega_i \text{ or } \omega_i X_j) \\ &\equiv M(y_i) + (\text{terms of degree } \geq 3). \end{aligned}$$

Since  $F_S$  is topologically generated by  $x_j$ 's, this proves (i).

**Proof of (ii):** Suppose that  $x_i$  is replaced by  $x_i^* = x_j x_i x_j^{-1}$ . As  $x_i = x_j^{-1} x_i^* x_j$ , we have  $1 + X_i =$

$(1 - X_j + X_j^2 - \cdots)(1 + X_i^*)(1 + X_j)$ . Therefore,  $X_i = X_i^* + (\text{terms containing } X_j X_i^* \text{ or } X_i^* X_j)$ . Each time  $X_i$  occurs in the Magnus expansion  $M(y_i)$ , it is to be replaced by this last expansion. The terms in the bracket give rise to terms of degree  $\geq 3$  in the new Magnus expansion of  $y_i$  using  $X_i^*$ . Hence, the new expansion has the same coefficient in degree 2 as the old. This completes the proof of (ii).

**Proof of (iii):** The identity  $M(x_j y_j - y_j x_j) = X_j \omega_j - \omega_j X_j$  implies

$$\begin{aligned} M([x_j, y_j]) &= M(1 + (x_j y_j - y_j x_j) x_j^{-1} y_j^{-1}) \\ &= 1 + (X_j \omega_j - \omega_j X_j)(1 - X_j + X_j^2 - \cdots) M(y_j^{-1}) \\ &= 1 \bmod M(I_{\mathbb{F}_2[[F_S]]}^3). \end{aligned}$$

In addition, we have  $M(x_j^{\text{Np}_j-1}) = (1 + X_j)^{\text{Np}_j-1} \equiv 1 \bmod M(I_{\mathbb{F}_2[[F_S]]}^3)$ . Hence, we have

$$x_j^{\text{Np}_j-1} [x_j, y_j] \equiv 1 \bmod I_{\mathbb{F}_2[[F_S]]}^3.$$

Moreover, by Proposition 1.4,  $x_{\infty_2} \in \Phi(G_S(2)) = G_S(2)^{(2)} \subset F_S^{(2)}$  and  $(F_S^{(2)})^2 \subset F_S^{(4)}$ ,  $x_{\infty_2}^2 \in F_S^{(4)}$ . Hence we have

$$M(x_{\infty_2}^2) = 1 + (\text{terms of degree } \geq 4),$$

which leads to the assertion.  $\square$

By Theorem 2.3, we call  $\mu_2(I)$  the mod 2 *arithmetic triple Milnor invariant* of  $S$  for the multi-index  $|I| = 3$ , when  $\left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1$  for  $\mathfrak{p}_i \in S$ ,  $\mu_2(J) = 0$  for any multi-index  $J$  of length  $\leq 2$  and  $\text{Np}_i \equiv 1 \bmod 4$  for  $\mathfrak{p}_i \in S$ .

A meaning of the mod 2 arithmetic triple Milnor invariants is given as follows. Let  $\overline{S} = S \cup S_{\mathbb{R}} = \{\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3\} \cup \{\infty_1, \infty_2\}$  be a finite set of primes of  $k$ , where  $\text{Np}_i \equiv 1 \bmod 4$  ( $1 \leq i \leq 3$ ) and  $\infty_1, \infty_2$  are real primes. We assume that  $\mu_2(ij) = 0$  for any multi-index of length 2. By Proposition 2.2, it means  $\left(\frac{\pi_i}{\pi_j}\right) = 1$  for  $i \neq j$ . Let  $N_3(\mathbb{F}_2)$  denote the group of 3 by 3 upper-triangular unipotent matrices over  $\mathbb{F}_2$ . We note that  $N_3(\mathbb{F}_2)$  is isomorphic to the dihedral group  $D_8 = \langle s, t \mid s^2 = t^4 = 1, sts^{-1} = t^{-1} \rangle$  of order 8 by the correspondence

$$s \mapsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad t \mapsto \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

We define the map

$$\rho : F_S \longrightarrow N_3(\mathbb{F}_2)$$

by

$$\rho(f) := \begin{pmatrix} 1 & \mu_2((1); f) & \mu_2((12); f) \\ 0 & 1 & \mu_2((2); f) \\ 0 & 0 & 1 \end{pmatrix}$$

for  $f \in F_S$ . By Lemma 2.1 (i), we see  $\rho$  is a group homomorphism. Let  $S' = \{\mathfrak{p}_1, \mathfrak{p}_2\}$  and let  $\overline{S'} = S' \cup S'_{\mathbb{R}}$  be the subset of  $S$  and  $\overline{S}$ .

**Theorem 2.4.** *Notations being as above, we assume that  $\left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1$  for  $\mathfrak{p}_i \in S$ .*

(1) *The homomorphism  $\rho$  is surjective and factors through the Galois group  $G_S(2)$  and  $G_{S'}(2)$ .*  
 (2) *Let  $K_\rho$  be the extension over  $k$  corresponding to  $\text{Ker}(\rho)$ . Then  $K_\rho$  is a Galois extension of  $k$  unramified outside  $\overline{S'}$  with Galois group  $\text{Gal}(K_\rho/k) = N_3(\mathbb{F}_2)$ , and each ramification index over  $\mathfrak{p}_i$  is 2.*

(3) *Assume that  $\left(\frac{\pi_i}{\pi_j}\right) = 1$  for  $i \neq j$ . For a Frobenius automorphism  $\sigma_3$  over  $\mathfrak{p}_3$ , we have*

$$\rho(\sigma_3) = \begin{pmatrix} 1 & 0 & \mu_2(123) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

In particular, we have

$$\mu_2(123) = 0 \iff \mathfrak{p}_3 \text{ is completely decomposed in } K_\rho/k.$$

**Proof.** (1) Since

$$\rho(x_1) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \rho(x_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

and so  $\rho(x_1), \rho(x_2)$  generate  $N_3(\mathbb{F}_2)$ ,  $\rho$  is surjective.

Next, by presentation of  $G_S(2)$  and  $G_{S'}(2)$ , we need to show

$$\rho(x_i^{\text{Np}_i-1}[x_i, y_i]) = E_3 (1 \leq i \leq 3) \text{ and } \rho(x_{\infty_2}^2) = E_3,$$

where  $E_3$  is the identity matrix of size 3. Since we can prove

$$\mu_2((1); x_i^{\text{Np}_i-1}[x_i, y_i]) = \mu_2((2); x_i^{\text{Np}_i-1}[x_i, y_i]) = \mu_2((12); x_i^{\text{Np}_i-1}[x_i, y_i]) = 0$$

and

$$\mu_2((1); x_{\infty_2}^2) = \mu_2((2); x_{\infty_2}^2) = \mu_2((12); x_{\infty_2}^2) = 0$$

as in the proof of Theorem 2.3, we are done.

(2) By (1), we obtain  $\text{Gal}(K_\rho/k) = N_3(\mathbb{F}_2)$ . Since  $\rho(x_3) = E_3$ ,  $K_\rho/k$  is unramified outside  $\overline{S'}$ . The ramification index over  $\mathfrak{p}_i$  is the order of  $\rho(x_i)$ , which is 2.

(3) For  $J$  with  $|J| = 1$ ,  $\mu_2(J; y_3) = 0$  by definition and the assumption. Hence the assertion follows.  $\square$

**Definition 2.5.** Let  $\mathfrak{p}_1, \mathfrak{p}_2$  be distinct finite primes of  $k$  such that  $\text{Np}_i \equiv 1 \pmod{4}$ . We call an extension  $K$  of  $k$  a *Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$*  if  $K/k$  is a Galois extension of  $k$  which is unramified outside  $\{\mathfrak{p}_1, \mathfrak{p}_2, \infty_1, \infty_2\}$  with ramification index of each  $\mathfrak{p}_i$  being 2 and whose Galois group is isomorphic to the group  $D_8 \cong N_3(\mathbb{F}_2)$ .

**Theorem 2.6.** Notations being as above, let  $\mathfrak{p}_1 = (\pi_1), \mathfrak{p}_2 = (\pi_2)$  be distinct finite primes of  $k$  such that  $\text{Np}_i \equiv 1 \pmod{4}$ . Suppose that  $\left(\frac{\pi_1}{\pi_2}\right) = \left(\frac{\pi_2}{\pi_1}\right) = 1$ ,  $\iota_{\infty_2}(\pi_1) > 0$ , and  $\iota_{\infty_2}(\pi_2) > 0$

(1) (Uniqueness) The Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  is unique if it exists.

(2) (Existence) We assume further  $\left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1$  for  $\mathfrak{p}_i$  for  $i = 1, 2$ . Then the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  exists.

**Proof.** (1) This is proved in a manner similar to the proof of [AMM, Theorem 4.1], [Miz, Proposition 3.7]. Let  $K/k$  be a Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  and  $S' = \{\mathfrak{p}_1 = (\pi_1), \mathfrak{p}_2 = (\pi_2)\}$ . We fix a presentation of  $G_{S'}(2)$ :

$$G_{S'}(2) = F/R,$$

where  $F$  is the free pro-2 group on the words  $x_1, x_2$  and  $R$  is a normal subgroup of  $F$  generated by  $x_1^{\text{Np}_1-1}[x_1, y_1], x_2^{\text{Np}_2-1}[x_2, y_2], x_{\infty_2}^2$ . Let  $N$  be the normal subgroup of  $F$  generated by  $x_1^2, x_2^2$ . Since any element of  $F^2$  can be expressed as a product of  $x_1^2$  and  $x_2^2$  modulo  $[F, F]$ ,  $N[F, F] = F^2[F, F]$ . By definition of  $R$ ,  $R$  is subgroup of  $N[F, F] = F^2[F, F]$ . By Proof of Proposition 2.2, our assumption  $\left(\frac{\pi_1}{\pi_2}\right) = \left(\frac{\pi_2}{\pi_1}\right) = 1$  implies that  $y_1, y_2 \in [F, F] \subset N[F, F]$ , so  $[x_1, y_1], [x_2, y_2] \in N[[F, F], F]$ . In addition, since  $N[F, F]$  corresponds to  $k(\sqrt{\pi_1}, \sqrt{\pi_2})$ , we have  $x_{\infty_2} \in N[F, F]$  by our assumption  $\iota_{\infty_2}(\pi_1) > 0$  and  $\iota_{\infty_2}(\pi_2) > 0$ . In otherwords,  $x_{\infty_2}^2 \in (N[F, F])^2 \subset N[[F, F], F]$ . Since

$$[x_1, x_2]^2 \equiv [x_1^2, x_2] \equiv 1 \pmod{N[[F, F], F]},$$

$N[F, F]/N[[F, F], F]$  is a cyclic group generated by  $[x_1, x_2]N[[F, F], F]$  of order at most 2.

Let  $\psi : G_{S'}(2) = F/R \longrightarrow \text{Gal}(K/k)$  be the natural homomorphism. Since  $[[D_8, D_8], D_8] = 1$  and ramification index of each  $\mathfrak{p}_i$  is 2,  $N[[F, F], F] \subset \text{Ker}(\psi)$ . Hence  $\psi$  induces an isomorphism

$$F/N[[F, F], F] \xrightarrow{\sim} \text{Gal}(K/k).$$

Therefore,  $K$  is uniquely determined as a fixed field of  $N[[F, F], F]/R$ .

(2) This follows from Theorem 2.4 (2).  $\square$

Suggested by Proposition 2.2, we may define the triple quadratic residue symbol  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  as follows.

**Definition 2.7.** Let  $\mathfrak{p}_1 = (\pi_1)$ ,  $\mathfrak{p}_2 = (\pi_2)$  and  $\mathfrak{p}_3 = (\pi_3)$  be distinct finite primes of  $k$  such that  $N\mathfrak{p}_i \equiv 1 \pmod{4}$ . Assume that the narrow class group of  $k$  is trivial and that

$$\left(\frac{\pi_1}{\pi_2}\right) = \left(\frac{\pi_2}{\pi_1}\right) = 1 \text{ and } \left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1 \ (1 \leq i \leq 3).$$

Then the triple quadratic residue symbol  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  is defined by

$$[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] := (-1)^{\mu_2(123)}.$$

**Theorem 2.8.** *Let the assumptions be as in Definition 2.7. Let  $K$  be the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  by Theorem 2.6. Then we have the following:*

$$[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] = 1 \iff \mathfrak{p}_3 \text{ is completely decomposed in } K/k.$$

**Proof.** The existence of a Rédei type  $D_8$ -extension  $K$  of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  follows from Theorem 2.4 (2). The uniqueness of  $K$  follows from Theorem 2.6. The assertion for  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  follows from Theorem 2.4 (3).  $\square$

**Corollary 2.9.** *Let the assumptions be as in Definition 2.7. Then, the following equation holds:*

$$[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] = [\mathfrak{p}_2, \mathfrak{p}_1, \mathfrak{p}_3].$$

**Proof.** The construction of Rédei type  $D_8$ -extension  $K$  of  $k$  depends on the set  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  of two primes, and is independent of the ordering of them. Therefore, by Theorem 2.8, the values of the two symbols  $[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3]$  and  $[\mathfrak{p}_2, \mathfrak{p}_1, \mathfrak{p}_3]$  are equal.  $\square$

### 3. RÉDEI TYPE $D_8$ -EXTENSIONS OVER REAL QUADRATIC FIELDS

In this section, we give examples of Rédei type  $D_8$ -extensions of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  in Theorem 2.6. We keep the same notations and assumptions as in the Section 2. Namely, let  $k$  be a real quadratic field whose narrow class number is one. By Gauss' genus theory ([O, Remark 4.7, p.172]), there is a prime number  $p = 2$  or  $p \equiv 1 \pmod{4}$  such that  $k = \mathbb{Q}(\sqrt{p})$ . We assume that  $p \equiv 1 \pmod{4}$ . Let  $\mathfrak{p}_1 = (\pi_1)$ ,  $\mathfrak{p}_2 = (\pi_2)$ ,  $\mathfrak{p}_3 = (\pi_3)$  be distinct finite primes of  $k$  such that  $N\mathfrak{p}_i \equiv 1 \pmod{4}$ ,  $\left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1$  ( $1 \leq i \leq 3$ ) and  $\left(\frac{\pi_i}{\pi_j}\right) = 1$  ( $1 \leq i \neq j \leq 3$ ). Let  $p_i$  denote the rational prime lying below  $\mathfrak{p}_i$ .

**Proposition 3.1.** *Suppose that both  $p_1$  and  $p_2$  are inert in  $k/\mathbb{Q}$  and  $p_1, p_2 \equiv 1 \pmod{4}$ . Let  $R$  be Rédei's extension of  $\mathbb{Q}$  for  $\{p_1, p_2\}$ , which is given by*

$$R = \mathbb{Q}(\sqrt{p_1}, \sqrt{p_2}, \sqrt{x + y\sqrt{p_1}}),$$

where  $x, y$  are non-zero integers satisfying

$$x^2 - p_1 y^2 - p_2 z^2 = 0, \quad y \equiv 0 \pmod{2}, \quad x - y \equiv 1 \pmod{4}$$

with some non-zero integer  $z$  ([R]). Then, the Rédei type  $D_8$ -extension  $K$  of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  is the composite field

$$K = R \cdot k = k(\sqrt{p_1}, \sqrt{p_2}, \sqrt{x + y\sqrt{p_1}}).$$

**Proof.** Recall that  $R$  is a Galois extension of  $\mathbb{Q}$  unramified outside  $\{p_1, p_2, \infty\}$  with the ramification index of each  $p_i$  being 2 and whose Galois group is isomorphic to the group  $D_8$  ([A], [R]). Since only  $\mathbb{Q}(\sqrt{p_1})$ ,  $\mathbb{Q}(\sqrt{p_2})$ ,  $\mathbb{Q}(\sqrt{p_1 p_2})$  are the intermediate quadratic fields of  $R/\mathbb{Q}$ ,  $R \cap k$  is  $\mathbb{Q}$ . By Galois theory,  $K/k$  is a Galois extension and  $\text{Gal}(K/k) = D_8$ . By Hilbert ramification theory,  $K/k$  is unramified outside  $\{\mathfrak{p}_1, \mathfrak{p}_2, \infty_1, \infty_2\}$ . Moreover, since  $p_1$  and  $p_2$  are inert in  $k/\mathbb{Q}$ ,

the ramification index of each  $\mathfrak{p}_i$  is 2. Therefore  $K$  is the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$ .  $\square$

Next, we consider the case that  $p_1$  or  $p_2$  is decomposed in  $k/\mathbb{Q}$ . To deal with this case, we recall some lemmas regarding ramification.

**Lemma 3.2** ([N] Kapitel III, (2.5) Theorem, p.209). *Let  $L/F$  be an extension of algebraic number fields,  $d(L/F)$  be the relative discriminant of  $L/F$ , and  $d(\alpha, L/F)$  be the relative discriminant of  $\alpha \in \mathcal{O}_L$  for  $L/F$ . Then  $d(L/F) \mid (d(\alpha, L/F))$ .*

Using Lemma 3.2, we obtain the following lemma.

**Lemma 3.3.** *Let  $k$  be a real quadratic field  $k = \mathbb{Q}(\sqrt{p})$ ,  $\pi$  be a prime element of  $k$  such that  $\pi \equiv 1 \pmod{4\mathcal{O}_k}$ . Then the extension  $k(\sqrt{\pi})/k$  is ramified at only  $(\pi)$ .*

**Proof.** We set  $\alpha_1 := (1 + \sqrt{\pi})/2$ . Since  $\alpha_1$  satisfies  $\alpha_1^2 - \alpha_1 + (1 - \pi)/4 = 0$  and  $\pi \equiv 1 \pmod{4\mathcal{O}_k}$ , we have  $\alpha_1 \in \mathcal{O}_{k(\sqrt{\pi})}$ . Then the relative discriminant of  $\alpha_1$  in  $k(\sqrt{\pi})/k$  is

$$d(\alpha_1, k(\sqrt{\pi})/k) = \begin{vmatrix} 1 & \alpha_1 \\ 1 & \overline{\alpha_1} \end{vmatrix}^2 = \pi.$$

By Lemma 3.2,  $k(\sqrt{\pi})/k$  can be ramified at only  $(\pi)$ .  $\square$

**Lemma 3.4.** *Let  $k$  be a real quadratic field with trivial narrow class group, and let  $\pi > 0$  be a prime element of  $\mathcal{O}_k$ , and let  $\varepsilon$  be the fundamental unit of  $k$ . We assume  $\pi \equiv 1 \pmod{4\mathcal{O}_k}$ . Then, the following are equivalent.*

- (i)  $\left(\frac{\varepsilon}{\pi}\right) = 1$ .
- (ii)  $\iota_{\infty_2}(\pi) > 0$ , where  $\iota_{\infty_2}$  is the conjugate embedding.

**Proof.** We calculate the Hilbert symbol  $\left(\frac{\varepsilon, \pi}{\mathfrak{p}}\right)$  for all primes  $\mathfrak{p}$  of  $k$  to use product formula. By definition of the Hilbert symbol, we have

$$\left(\frac{\varepsilon, \pi}{\mathfrak{p}}\right) = \frac{(\varepsilon, k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}})(\sqrt{\pi})}{\sqrt{\pi}},$$

where  $(\ , k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}}) : k_{\mathfrak{p}}^{\times} \rightarrow \text{Gal}(k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}})$  is a norm residue symbol. Case  $\mathfrak{p} \nmid (\pi)\infty$ . By Lemma 3.3,  $k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}}$  is an unramified extension. So, the norm residue symbol factors through  $\text{Gal}(k_{\mathfrak{p}}^{\text{ur}}/k_{\mathfrak{p}})$  ;

$$k_{\mathfrak{p}}^{\times} \xrightarrow{(\ , k_{\mathfrak{p}}^{\text{ur}}/k_{\mathfrak{p}})} \text{Gal}(k_{\mathfrak{p}}^{\text{ur}}/k_{\mathfrak{p}}) \xrightarrow{\text{res}} \text{Gal}(k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}}).$$

Since the local class field theory, the kernel of  $(\ , k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}})$  is  $\mathcal{O}_{\mathfrak{p}}^{\times}$ , and  $\varepsilon \in \mathcal{O}_{\mathfrak{p}}^{\times}$ , so  $(\varepsilon, k_{\mathfrak{p}}^{\text{ur}}/k_{\mathfrak{p}}) = \text{id}_{k_{\mathfrak{p}}^{\text{ur}}}$ . Therefore  $(\varepsilon, k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}}) = \text{id}_{k_{\mathfrak{p}}(\sqrt{\pi})}$  and hence we have  $\left(\frac{\varepsilon, \pi}{\mathfrak{p}}\right) = 1$ .

Case  $\mathfrak{p} = (\pi)$ . By definition of the quadratic residue symbol,  $\left(\frac{\varepsilon, \pi}{\mathfrak{p}}\right) = \left(\frac{\varepsilon}{\pi}\right)$ .

Case  $\mathfrak{p} \mid \infty$ . Since  $\mathfrak{p}$  is a real prime,  $k_{\mathfrak{p}} = \mathbb{R}$ . If  $\mathfrak{p} = \infty_1$ ,  $(\varepsilon, k_{\mathfrak{p}}(\sqrt{\pi})/k_{\mathfrak{p}}) = (\varepsilon, \mathbb{R}/\mathbb{R}) = \text{id}_{\mathbb{R}}$  since  $\pi > 0$ . Thus, we have  $\left(\frac{\varepsilon, \pi}{\mathfrak{p}}\right) = 1$ . If  $\mathfrak{p} = \infty_2$ , since  $(\iota_{\infty_2}(\varepsilon), k_{\mathfrak{p}}(\sqrt{\iota_{\infty_2}(\pi)})/k_{\mathfrak{p}}) = (\iota_{\infty_2}(\varepsilon), \mathbb{R}(\sqrt{\iota_{\infty_2}(\pi)})/\mathbb{R})$ , then

$$\left(\frac{\varepsilon, \pi}{\mathfrak{p}}\right) = \begin{cases} 1 & \text{if } \iota_{\infty_2}(\pi) > 0 \\ -1 & \text{if } \iota_{\infty_2}(\pi) < 0. \end{cases}$$

From the above calculation and Hilbert's product formula, we have

$$\left(\frac{\varepsilon}{\pi}\right) = 1 \iff \iota_{\infty_2}(\pi) > 0.$$

$\square$

**Theorem 3.5.** Let  $k = \mathbb{Q}(\sqrt{p})$  be a real quadratic field with trivial narrow class group and we assume  $p \equiv 5 \pmod{8}$ . Let  $\varepsilon$  be the fundamental unit of  $k$ , and  $\mathfrak{p}_1 = (\pi_1), \mathfrak{p}_2 = (\pi_2)$  are prime ideals of  $\mathcal{O}_k$  such that:

$$\left(\frac{\varepsilon}{\pi_i}\right) = 1, \quad \left(\frac{\pi_i}{\pi_j}\right) = 1, \quad \pi_i \equiv 1 \pmod{4\mathcal{O}_k}, \quad \pi_i > 0 \quad (1 \leq i \neq j \leq 2).$$

Then the followings hold:

- (1) There is a non-zero  $\mathcal{O}_k$  solution  $(x, y, z)$  of  $x^2 - \pi_1 y^2 - \pi_2 z^2 = 0$  such that  $\gcd(x, y, z) = 1$ ,  $y \in 2\mathcal{O}_k$ .
- (2) If  $x - y \equiv 1 \pmod{4\mathcal{O}_k}$ , the Rédei type  $D_8$ -extension  $K$  of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$  is given by

$$K = k(\sqrt{\pi_1}, \sqrt{\pi_2}, \sqrt{x + y\sqrt{\pi_1}}).$$

**Proof.** (1) We use Hasse–Minkowski theorem to show the equation in (1) has a nontrivial  $\mathcal{O}_k$  solution. It suffices to show that the Hilbert symbol  $\left(\frac{\pi_1, \pi_2}{\mathfrak{p}}\right)$  is 1 for all primes  $\mathfrak{p}$  of  $k$ .

Case  $\mathfrak{p} \nmid \mathfrak{p}_1 \mathfrak{p}_2 \infty$ . Since  $\pi_2 \in \mathcal{O}_{\mathfrak{p}}^\times$  and Lemma 3.3,  $k_{\mathfrak{p}}(\sqrt{\pi_2})/k_{\mathfrak{p}}$  is an unramified extension. Furthermore, like the proof of Lemma 3.4, it can be seen that  $\pi_1 \in \mathcal{O}_{\mathfrak{p}}^\times$ ,  $(\pi_1, k_{\mathfrak{p}}(\sqrt{\pi_2})/k_{\mathfrak{p}}) = \text{id}_{k_{\mathfrak{p}}(\sqrt{\pi_2})}$ , and hence we have  $\left(\frac{\pi_1, \pi_2}{\mathfrak{p}}\right) = 1$ .

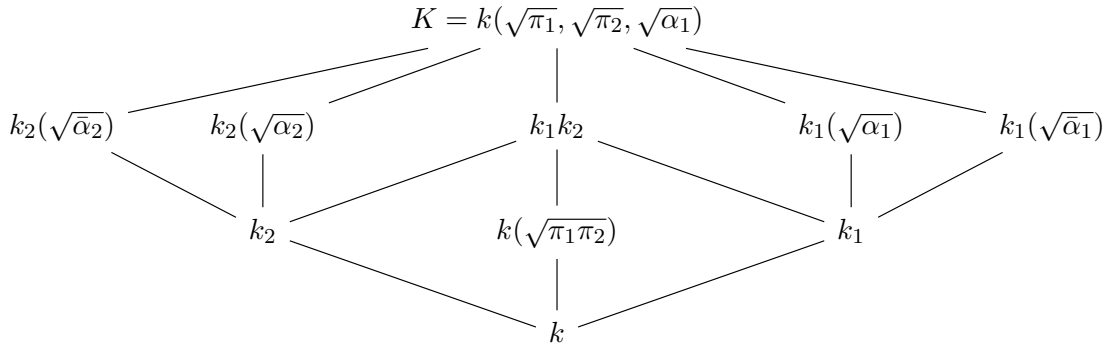
Case  $\mathfrak{p} \mid \mathfrak{p}_1 \mathfrak{p}_2$ . Since the quadratic residue symbols  $\left(\frac{\pi_2}{\pi_1}\right) = \left(\frac{\pi_1}{\pi_2}\right) = 1$ , we have  $\left(\frac{\pi_1, \pi_2}{\mathfrak{p}}\right) = 1$ .

Case  $\mathfrak{p} \mid \infty$ . Since  $\mathfrak{p}$  is a real prime,  $k_{\mathfrak{p}} = \mathbb{R}$ . If  $\mathfrak{p} = \infty_1$ ,  $(\pi_1, k_{\mathfrak{p}}(\sqrt{\pi_2})/k_{\mathfrak{p}}) = (\pi_1, \mathbb{R}(\sqrt{\pi_2})/\mathbb{R}) = (\pi_1, \mathbb{R}/\mathbb{R}) = 1$  and hence we have  $\left(\frac{\pi_1, \pi_2}{\mathfrak{p}}\right) = 1$ .

If  $\mathfrak{p} = \infty_2$ ,  $\iota_{\infty_2}(\pi_2) > 0$  by Lemma 3.4. Therefore  $(\pi_1, k_{\mathfrak{p}}(\sqrt{\iota_{\infty_2}(\pi_2)})/k_{\mathfrak{p}}) = (\pi_1, \mathbb{R}(\sqrt{\iota_{\infty_2}(\pi_2)})/\mathbb{R}) = (\pi_1, \mathbb{R}/\mathbb{R}) = 1$  and hence we have  $\left(\frac{\pi_1, \pi_2}{\mathfrak{p}}\right) = 1$ .

From the above, the Hilbert symbol  $\left(\frac{\pi_1, \pi_2}{\mathfrak{p}}\right)$  is 1 for all primes  $\mathfrak{p}$  of  $k$ . Then there is a non-zero  $\mathcal{O}_k$  solution of  $x^2 - \pi_1 y^2 - \pi_2 z^2 = 0$  by Hasse–Minkowski theorem. Since the class number of  $k$  is one, we can assume  $\gcd(x, y, z) = 1$ . Since  $x^2 - \pi_1 y^2 - \pi_2 z^2 = 0$ , we have  $x^2 \equiv y^2 + z^2 \pmod{4\mathcal{O}_k}$ . If  $x^2 \equiv 0 \pmod{4\mathcal{O}_k}$ , then it causes a contradiction to  $\gcd(x, y, z) = 1$ . Therefore we can set  $y^2 \equiv 0 \pmod{4\mathcal{O}_k}$ . This means that  $y \in 2\mathcal{O}_k$ .

(2) First, we easily see that  $K = k(\sqrt{\pi_1}, \sqrt{\pi_2}, \sqrt{x + y\sqrt{\pi_1}})$  is a  $D_8$ -extension of  $k$ , where  $\alpha_1 := x + y\sqrt{\pi_1}$ . In fact, to be precise, the intermediate fields of  $K$  over  $k$  are given in the following, where  $k_i = k(\sqrt{\pi_i})$ , and  $\alpha_2 = (\sqrt{\alpha_1} + \sqrt{\alpha_1})^2$ .



Next, let us see the ramifications in  $K/k$ . We prove that  $k_1(\sqrt{\alpha_1})/k_1$  is ramified at only primes dividing  $\pi_2$ . We first claim that  $\theta := \frac{1+\sqrt{\alpha_1}}{2} \in \mathcal{O}_{k_1(\sqrt{\alpha_1})}$ . We set  $\beta := \frac{1-\alpha_1}{4}$ . It can be seen that  $\beta$  satisfies  $\beta^2 - (\frac{1-(x-y)}{2} - \frac{y}{2})\beta + \frac{1}{16}((1-x)^2 - y^2 + (1-\pi_1)y^2) = 0$ . Since  $x - y \equiv 1$

mod  $4\mathcal{O}_k$  and  $y \in 2\mathcal{O}_k$ ,  $1 - x + y \in 4\mathcal{O}_k$  and  $1 - x - y = 1 - x + y - 2y \in 4\mathcal{O}_k$ . Then  $(1 - x)^2 - y^2 = (1 - x + y)(1 - x - y) \in 16\mathcal{O}_k$ . Since  $\pi_1 \equiv 1 \pmod{4\mathcal{O}_k}$  and  $y \in 2\mathcal{O}_k$ ,  $(1 - \pi_1)y^2 \in 16\mathcal{O}_k$ . Therefore,  $\beta \in \mathcal{O}_{k_1}$ . Since  $\theta$  satisfies  $\theta^2 - \theta + \beta = 0$ ,  $\theta \in \mathcal{O}_{k_1(\sqrt{\alpha_1})}$ . Since the relative discriminant of  $\theta$  in  $k_1(\sqrt{\alpha_1})/k_1$  is

$$d(\theta, k_1(\sqrt{\alpha_1})/k_1) = \begin{vmatrix} 1 & \theta \\ 1 & \bar{\theta} \end{vmatrix}^2 = \alpha_1.$$

Since  $N_{k_1(\sqrt{\alpha_1})/k_1}(\alpha_1) = \pi_2$  and Lemma 3.2,  $k_1(\sqrt{\alpha_1})/k_1$  is ramified at only primes dividing  $\pi_2$ .  $\square$

Here is a numerical example of Theorem 3.5.

**Example 3.6.** Let  $k = \mathbb{Q}(\sqrt{5})$ ,  $\pi_1 = 33 + 8\sqrt{5}$  ( $N_{k/\mathbb{Q}}(\pi_1) = 769$ ),  $\pi_2 = 17$ ,  $\pi_3 = \frac{23+5\sqrt{5}}{2}$  ( $N_{k/\mathbb{Q}}(\pi_3) = 101$ ). We set  $\mathfrak{p}_i = (\pi_i)$  ( $1 \leq i \leq 3$ ). Then we have

$$\begin{aligned} \left(\frac{\varepsilon}{\pi_i}\right) &= 1, \quad \left(\frac{\pi_i}{\pi_j}\right) = 1, \quad \pi_i \equiv 1 \pmod{4\mathcal{O}_k}, \quad \pi_i > 0 \quad (1 \leq i \neq j \leq 2), \\ \left(\frac{\varepsilon}{\pi_3}\right) &= 1, \quad \left(\frac{\pi_i}{\pi_j}\right) = 1, \quad N\mathfrak{p}_3 \equiv 1 \pmod{4} \quad (1 \leq i \neq j \leq 3), \end{aligned}$$

where  $\varepsilon = (1 + \sqrt{5})/2$ . It is easy to see that

$$(-23 - 14\sqrt{5})^2 - \pi_1^2 - \pi_2(2 + 3\sqrt{5})^2 = 0 \text{ and } (-23 - 14\sqrt{5}) - 2 \equiv 1 \pmod{4\mathcal{O}_k}.$$

Therefore,  $K = k(\sqrt{\pi_1}, \sqrt{\pi_2}, \sqrt{\alpha_1})$  is the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$ , where  $\alpha_1 = (-23 - 14\sqrt{5}) + 2\sqrt{\pi_1}$ . Since  $\mathfrak{p}_3$  is not completely decomposed in  $K$ , we have

$$[\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3] = -1.$$

**Remark 3.7.** From the viewpoint of arithmetic topology, mod 2 arithmetic Milnor invariants may be regarded as an arithmetic analogue of (higher) linking numbers of a link. So, the triple of primes  $(\mathfrak{p}_1 = (33 + 8\sqrt{5}), \mathfrak{p}_2 = (17), \mathfrak{p}_3 = (\frac{23+5\sqrt{5}}{2}))$  in Example 3.6 may be called the *Borromean primes*.

Next, we give other numerical examples, which are not contained in the cases of Proposition 3.1 and Theorem 3.5.

**Example 3.8.** The case that  $p_1$  is decomposed and  $p_2$  is inert in  $k/\mathbb{Q}$ . We give an example for  $k = \mathbb{Q}(\sqrt{5})$ ,  $p_1 = 29$  and  $p_2 = 13$ . We take  $\pi_1 = (11 + \sqrt{5})/2$  and  $\pi_2 = 13$ . Then  $\left(\frac{\pi_1}{\pi_2}\right) = \left(\frac{\pi_2}{\pi_1}\right) = 1$  and  $\left(\frac{\varepsilon}{\mathfrak{p}_1}\right) = \left(\frac{\varepsilon}{\mathfrak{p}_2}\right) = 1$ , where  $\varepsilon = (1 + \sqrt{5})/2$ . In this case,  $K = k(\sqrt{\pi_1}, \sqrt{\pi_2}, \sqrt{\alpha_1})$  is the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$ , where  $\alpha_1 = (-1 - 9\sqrt{5} + 6\sqrt{\pi_1})/4$ .

**Proof.** Let  $k_1 := k(\sqrt{\pi_1})$ . We prove that  $k_1/k$  can be ramified at only over  $\mathfrak{p}_1$  and  $k_1(\sqrt{\alpha_1})/k_1$  is ramified at only primes dividing  $\pi_2$ . First we set  $\lambda_1 := (1 + \sqrt{5} + 2\sqrt{\pi_1})/4$ . Since  $\lambda_1$  satisfies  $\lambda_1^2 - \frac{1+\sqrt{5}}{2}\lambda_1 - 1 = 0$ ,  $\lambda_1 \in \mathcal{O}_{k_1}$ . Since the relative discriminant of  $\lambda_1$  in  $k_1/k$  is

$$d(\lambda_1, k_1/k) = \begin{vmatrix} 1 & \lambda_1 \\ 1 & \bar{\lambda}_1 \end{vmatrix}^2 = \pi_1.$$

By Lemma 3.2,  $k_1/k$  can be ramified at only over  $\mathfrak{p}_1$ . Next we set  $\lambda_2 := ((\sqrt{5} - 1) - (\sqrt{5} + 1)\sqrt{\pi_1} + 4\sqrt{\alpha_1})/8$ . Since  $\lambda_2$  satisfies  $\lambda_2^8 + \lambda_2^7 - 2\lambda_2^6 + 3\lambda_2^5 + 11\lambda_2^4 - \lambda_2^3 - 18\lambda_2^2 - 13\lambda_2 - 1 = 0$ ,  $\lambda_2 \in \mathcal{O}_{k_1(\sqrt{\alpha_1})}$ . Since the relative discriminant of  $\lambda_2$  in  $k_1(\sqrt{\alpha_1})/k_1$  is

$$d(\lambda_2, k_1(\sqrt{\alpha_1})/k_1) = \begin{vmatrix} 1 & \lambda_2 \\ 1 & \bar{\lambda}_2 \end{vmatrix}^2 = \alpha_1.$$

Since  $N_{k_1/k}(\alpha_1) = \pi_2$  and Lemma 3.2,  $k_1(\sqrt{\alpha_1})/k_1$  is ramified at only primes dividing  $\pi_2$ . Therefore the extension  $K/k$  is the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$ .  $\square$

**Example 3.9.** The case that  $p_1$  and  $p_2$  are decomposed in  $k/\mathbb{Q}$ . We give an example  $k = \mathbb{Q}(\sqrt{5})$ ,  $p_1 = 29$  and  $p_2 = 89$ . We take  $\pi_1 = (11 + \sqrt{5})/2$  and  $\pi_2 = (19 + \sqrt{5})/2$ . Then  $\left(\frac{\pi_1}{\pi_2}\right) = \left(\frac{\pi_2}{\pi_1}\right) = 1$  and  $\left(\frac{\varepsilon}{p_1}\right) = \left(\frac{\varepsilon}{p_2}\right) = 1$ , where  $\varepsilon = (1 + \sqrt{5})/2$ . In this case,  $K = k(\sqrt{\pi_1}, \sqrt{\pi_2}, \sqrt{\alpha_1})$  is the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$ , where  $\alpha_1 = (6\sqrt{5} + (1 - \sqrt{5})\sqrt{\pi_1})/4$ .

**Proof.** Let  $k_1 := k(\sqrt{\pi_1})$  and  $k_2 := k(\sqrt{\pi_2})$ . We prove that  $k_2/k$  is ramified at only  $\mathfrak{p}_2$  and  $k_1(\sqrt{\alpha_1})/k_1$  is ramified at only primes dividing  $\pi_2$ . First we set  $\theta_1 := (1 + \sqrt{5} + 2\sqrt{\pi_2})/4$ . Since  $\theta_1$  satisfies  $\theta_1^2 - \frac{1+\sqrt{5}}{2}\theta_1 - 2 = 0$ ,  $\theta_1 \in \mathcal{O}_{k_2}$ . Since the relative discriminant of  $\theta_1$  in  $k_2/k$  is

$$d(\theta_1, k_2/k) = \begin{vmatrix} 1 & \theta_1 \\ 1 & \theta_1 \end{vmatrix}^2 = \pi_2.$$

By Lemma 3.2,  $k_2/k$  is ramified at only over  $\mathfrak{p}_2$ . Next we set  $\theta_2 := ((2\sqrt{5} + 4) - (\sqrt{5} + 3)\sqrt{\pi_1} + 4\sqrt{\alpha_1})/8$ . Since  $\theta_2$  satisfies  $\theta_2^8 - 4\theta_2^7 + 3\theta_2^5 - 2\theta_2^4 + 12\theta_2^3 + 2\theta_2^2 + 5\theta_2 - 1 = 0$ ,  $\theta_2 \in \mathcal{O}_{k_1(\sqrt{\alpha_1})}$ . Since the relative discriminant of  $\theta_2$  in  $k_1(\sqrt{\alpha_1})/k_1$  is

$$d(\theta_2, k_1(\sqrt{\alpha_1})/k_1) = \begin{vmatrix} 1 & \theta_2 \\ 1 & \theta_2 \end{vmatrix}^2 = \alpha_1.$$

Since  $N_{k_1/k}(\alpha_1) = \pi_2$  and Lemma 3.2,  $k_1(\sqrt{\alpha_1})/k_1$  is ramified at only primes dividing  $\pi_2$ . Therefore the extension  $K/k$  is the Rédei type  $D_8$ -extension of  $k$  for  $\{\mathfrak{p}_1, \mathfrak{p}_2\}$ .  $\square$

#### 4. MASSEY PRODUCTS IN GALOIS COHOMOLOGY

In this section, we interpret our mod 2 arithmetic Milnor invariants and triple quadratic residue symbols in terms of the Massey product in Galois cohomology. Our theorem is seen as a generalization of the known relation between the cup product and the quadratic residue symbol to the triple case, and also a generalization of the previous result by Morishita ([Mo3]) in the case of the rational number field to the real quadratic fields. It may be regarded as a mod 2 arithmetic analogue of the corresponding topological result due to Turaev ([Tu]).

Let  $G$  be a pro-2 group and let  $A$  be a commutative ring with identity on which  $G$  acts trivially. Let  $C^j(G, A)$  be the  $A$ -module of inhomogeneous  $j$ -cochains ( $j \geq 0$ ) of  $G$  with coefficients in  $A$  and we consider the differential graded algebra  $(C^\bullet(G, A), d)$ , where the product structure on  $C^\bullet(G, A) = \bigoplus_{j \geq 0} C^j(G, A)$  is given by the cup product and the differential  $d$  is the coboundary operator. Then we have the cohomology  $H^*(G, A) = H^*(C^\bullet(G, A))$  of the pro-2 group  $G$  with coefficients in  $A$ . In the following, we consider Massey products in  $H^1(G, A)$ . For the sign convention, we follow [D]. Let  $\chi_1, \dots, \chi_n \in H^1(G, A)$  ( $n \geq 2$ ). An  $n$ -th Massey product  $\langle \chi_1, \dots, \chi_n \rangle$  is said to be *defined* if there is an array

$$\Omega = \{\omega_{ij} \in C^1(G, A) \mid 1 \leq i < j \leq n+1, (i, j) \neq (1, n+1)\}$$

such that

$$\begin{cases} [\omega_{i, i+1}] = \chi_i & (1 \leq i \leq n), \\ d\omega_{ij} = \sum_{a=i+1}^{j-1} \omega_{ia} \cup \omega_{aj} & (j \neq i+1). \end{cases}$$

Such an array  $\Omega$  is called a *defining system* for  $\langle \chi_1, \dots, \chi_n \rangle$ . The value of  $\langle \chi_1, \dots, \chi_n \rangle$  relative to  $\Omega$ , denoted by  $\langle \chi_1, \dots, \chi_n \rangle_\Omega$ , is defined by the cohomology class represented by the 2-cocycle

$$\sum_{a=2}^n \omega_{1a} \cup \omega_{a, n+1}.$$

We define the Massey product  $\langle \chi_1, \dots, \chi_n \rangle$  to be the subset of  $H^2(G, A)$  consisting of elements  $\langle \chi_1, \dots, \chi_n \rangle_\Omega$  for some defining system  $\Omega$ . By convention,  $\langle \chi \rangle = 0$ . We recall the following basic fact (cf. [Kr]).

**Lemma 4.1.** *We have  $\langle \chi_1, \chi_2 \rangle = \chi_1 \cup \chi_2$ . For  $n \geq 3$ ,  $\langle \chi_1, \dots, \chi_n \rangle$  is defined and consists of a single element if  $\langle \chi_{j_1}, \dots, \chi_{j_a} \rangle = 0$  for all proper subsets  $\{j_1, \dots, j_a\}$  ( $a \geq 2$ ) of  $\{1, \dots, n\}$ . (In this case, we denote the single element by  $\langle \chi_1, \dots, \chi_n \rangle$ .)*

Next, we recall a relation between Massey products and Magnus coefficients. Suppose that  $G$  is a finitely generated pro-2 group with a minimal presentation

$$1 \longrightarrow N \longrightarrow F \xrightarrow{\psi} G \longrightarrow 1,$$

where  $F$  is a free pro-2 group on generators  $x_1, \dots, x_n$  with  $n = \dim_{\mathbb{F}_2} H^1(G, \mathbb{F}_2)$ . We set  $\tau_i := \psi(x_i)$  ( $1 \leq i \leq n$ ). We assume that  $\psi$  induces the isomorphism  $F/\Phi(F) \simeq G/\Phi(G)$  so that  $\psi$  induces the isomorphism  $\psi^* : H^1(G, \mathbb{F}_2) \simeq H^1(F, \mathbb{F}_2)$ . We let

$$\text{tra} : H^1(N, \mathbb{F}_2)^G \rightarrow H^2(G, \mathbb{F}_2)$$

be the transgression defined as follows. For  $a \in H^1(N, \mathbb{F}_2)^G$ , choose a 1-cochain  $b \in C^1(F, \mathbb{F}_2)$  such that  $b|_N = a$ . Since the value  $db(f_1, f_2)$ ,  $f_i \in F$ , depends only on the cosets  $f_i \bmod N$ ,  $db$  defines a 2-cocycle  $c$  of  $G$ . Then  $\text{tra}(a)$  is defined by the class of  $c$ . By the Hochschild-Serre spectral sequence,  $\text{tra}$  is an isomorphism and so we have the dual isomorphism, called the Hopf isomorphism,

$$\text{tra}^\vee : H_2(G, \mathbb{F}_2) \xrightarrow{\sim} H_1(N, \mathbb{F}_2)_G = N/N^2[N, F].$$

Then we have the following proposition (cf. [St, Lemma 1.5]). The proof goes in the same manner as in [Mo3, Theorem 2.2.2].

**Proposition 4.2.** *Notations being as above, let  $\chi_1, \dots, \chi_n \in H^1(G, \mathbb{F}_2)$  ( $n \geq 2$ ). Let  $f \in N$  and set  $\delta := (\text{tra}^\vee)^{-1}(f \bmod N^2[N, F])$ . Assume that all Massey products up to length  $n-1$  are trivial. Then  $N \subset F^{(n)}$  and we have*

$$\langle \chi_1, \dots, \chi_n \rangle(\delta) = \sum_{\substack{I=(i_1 \dots i_n) \\ 1 \leq i_1, \dots, i_n \leq n}} \chi_1(x_{i_1}) \cdots \chi_n(x_{i_n}) \mu_I(I; f).$$

Let us be back in our arithmetic situation and keep the same notations as in Section 2. So let  $k$  be the real quadratic field and let  $S := \{\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3\}$  be a set of all distinct primes where  $N\mathfrak{p}_i \equiv 1 \pmod{4}$  and  $\left(\frac{\varepsilon}{\mathfrak{p}_i}\right) = 1$ . Let  $\overline{S} := S \cup S_{\mathbb{R}}$  where  $S_{\mathbb{R}} = \{\infty_1, \infty_2\}$  be the set of real primes of  $k$ . By Theorem 1.3, we have the following minimal presentation of the Galois group  $G_S(2)$  of maximal pro-2 extension over  $k$  unramified outside  $\overline{S}$

$$G_S(2) = \left\langle x_1, x_2, x_3 \left| \begin{array}{l} x_1^{N\mathfrak{p}_1-1}[x_1, y_1] = x_2^{N\mathfrak{p}_2-1}[x_2, y_2] = x_3^{N\mathfrak{p}_3-1}[x_3, y_3] = 1 \\ x_{\infty_2}^2 = 1 \end{array} \right. \right\rangle \\ = F_S/N_S.$$

Here  $x_i$  denotes the word representing a monodromy  $\tau_i$  over  $\mathfrak{p}_i$  in  $k_S(2)/k$  ( $1 \leq i \leq 3$ ) and  $F_S$  denotes the free pro-2 group on  $x_1, x_2, x_3$ . The pro-2 word  $y_i$  represents a Frobenius automorphism  $\sigma_i$  over  $\mathfrak{p}_i$  in  $k_S(2)/k$  and  $N_S$  denotes the closed subgroup of  $F_S$  generated normally by  $x_i^{N\mathfrak{p}_i-1}[x_i, y_i]$  for  $1 \leq i \leq 3$  and  $x_{\infty_2}^2$ . We set  $\delta_i := (\text{tra}^\vee)^{-1}(x_i^{N\mathfrak{p}_i-1}[x_i, y_i])$ , where  $\text{tra}^\vee : H_2(G_S(2), \mathbb{F}_2) \xrightarrow{\sim} N_S/N_S^2[N_S, F_S]$  is the Hopf isomorphism. Let  $\chi_1, \chi_2, \chi_3 \in H^1(G_S(2), \mathbb{F}_2)$  be the Kronecker dual to the monodromies  $\tau_1, \tau_2, \tau_3$ , namely,  $\chi_i(\tau_j) = \delta_{i,j}$ .

**Theorem 4.3.** *Notations being as above, we have, for  $1 \leq i \neq j \leq 3$ ,*

$$\left(\frac{\pi_i}{\pi_j}\right) = (-1)^{\langle \chi_i, \chi_j \rangle(\delta_j)}, \quad \left(\frac{\pi_j}{\pi_i}\right) = (-1)^{\langle \chi_i, \chi_j \rangle(\delta_i)}.$$

**Proof.** By Proposition 4.2, we have

$$\langle \chi_i, \chi_j \rangle(\delta_a) = \mu_2((ij); x_a^{N\mathfrak{p}_a-1}[x_a, y_a]).$$

Here, noting that the Magnus coefficients are in  $\mathbb{F}_2$ , we have

$$\begin{aligned} M(x_a^{\mathbb{N}p_a-1}[x_a, y_a]) &= M(x_a^{\mathbb{N}p_a-1})M(x_a)M(y_a)M(x_a)^{-1}M(y_a)^{-1}, \\ M(x_a) &= 1 + X_a, M(x_a^{\mathbb{N}p_a-1}) = 1 + (\text{terms of degree } \geq 4), \\ M(x_a)^{-1} &= 1 + X_a + X_a^2 + (\text{terms of degree } \geq 3), \\ M(y_a) &= 1 + \sum_i \mu_2(ia)X_i + \sum_{i,j} \mu_2(ija)X_iX_j + (\text{terms of degree } \geq 3), \\ M(y_a)^{-1} &= 1 + \sum_i \mu_2(ia)X_i + \sum_{i,j} \{\mu_2(ija) + \mu_2(ia)\mu_2(ja)\}X_iX_j + (\text{terms of degree } \geq 3), \end{aligned}$$

The straightforward calculation yields

$$M(x_a^{\mathbb{N}p_a-1}[x_a, y_a]) = 1 + \sum_i \mu_2(ia)X_aX_i + \sum_i \mu_2(ia)X_iX_a + (\text{terms of degree } \geq 3).$$

Hence we obtain

$$\langle \chi_i, \chi_j \rangle (\delta_a) = \begin{cases} \mu_2(ij) & a = j, \\ \mu_2(ji) & a = i, \\ 0 & a \neq i, j. \end{cases}$$

The assertion follows from Theorem 2.2. □

We assume that

$$\langle \chi_i, \chi_j \rangle = 0 \quad (1 \leq i \neq j \leq 3),$$

which is equivalent to the condition

$$\left( \frac{\pi_i}{\pi_j} \right) = 1 \quad (1 \leq i \neq j \leq 3)$$

by Proposition 2.3, and the mod 2 Milnor invariants  $\mu_2(abc)$  ( $\{a, b, c\} = \{1, 2, 3\}$ ) are well defined. By the definition of Massey products and Lemma 4.1, there are 1-cochains  $\omega_{13}, \omega_{24} \in C^1(G_S(2), \mathbb{F}_2)$  such that

$$\langle \chi_1, \chi_2 \rangle = d\omega_{13}, \quad \langle \chi_2, \chi_3 \rangle = d\omega_{24},$$

and we have the triple Massey product  $\langle \chi_1, \chi_2, \chi_3 \rangle$  defined by

$$\langle \chi_1, \chi_2, \chi_3 \rangle = [\chi_1 \cup \omega_{24} + \omega_{13} \cup \chi_3].$$

**Theorem 4.4.** *Assume that*

$$\left( \frac{\pi_i}{\pi_j} \right) = 1 \quad (1 \leq i \neq j \leq 3).$$

*Then we have*

$$[\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3] = (-1)^{\langle \chi_1, \chi_2, \chi_3 \rangle (\delta_3)}, [\mathbf{p}_2, \mathbf{p}_3, \mathbf{p}_1] = (-1)^{\langle \chi_1, \chi_2, \chi_3 \rangle (\delta_1)}.$$

*In particular,  $\mu(123)$  and  $[\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3]$  are independent of choices of 1-cochains  $\omega_{13}$  and  $\omega_{24}$ .*

**Proof.** Noting by the assumption that all  $\mu_2(ij) = 0$ , we have

$$\begin{aligned} M(x_a) &= 1 + X_a, M(x_a^{\mathbb{N}p_a-1}) = 1 + (\text{terms of degree } \geq 4), \\ M(x_a)^{-1} &= 1 + X_a + X_a^2 + X_a^3 + (\text{terms of degree } \geq 4), \\ M(y_a) &= 1 + \sum_{i,j} \mu_2(ija)X_iX_j + \sum_{i,j,k} \mu_2(ijk)aX_iX_jX_k + (\text{terms of degree } \geq 4), \\ M(y_a)^{-1} &= 1 + \sum_{i,j} \mu_2(ija)X_iX_j + \sum_{i,j,k} \mu_2(ijk)aX_iX_jX_k + (\text{terms of degree } \geq 4). \end{aligned}$$

The straightforward calculation yields

$$M(x_a^{\mathbb{N}p_a-1}[x_a, y_a]) = 1 + \sum_{i,j} \mu_2(ija)X_aX_iX_j + \sum_{i,j} \mu_2(ija)X_iX_jX_a + (\text{terms of degree } \geq 4).$$

Hence we obtain

$$\langle \chi_1, \chi_2, \chi_3 \rangle(\delta_a) = \begin{cases} \mu_2(123) & a = 3, \\ \mu_2(231) & a = 1, \\ 0 & a \neq 1, 3. \end{cases}$$

The assertion follows from Definition 2.7.  $\square$

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